

Melvin Orichi Socana

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PROFESSIONAL SUMMARY

Data & Machine Learning Platform Engineer (Stanford MS, Electrical Engineering) specializing in Airflow orchestration, BigQuery/Vertex machine learning pipelines, and infrastructure modernization. Seeking roles building production ML systems and infrastructure.

EDUCATION

Stanford University, Stanford, CA

Graduated April 2024

Master of Science (MS): Electrical Engineering (Hardware & Software)

EXPERIENCE

Handshake, San Francisco, CA

January 2025 - Present

Machine Learning & Data Platform Engineer

- Engineered a centralized, company-wide LLM Orchestration Platform that democratized access to Generative AI, enabling all users to easily define, schedule, and execute large-scale or recurring LLM jobs.
- Increased Data Platform stability by 35% by leading the critical migration of complex Airflow DAGs (Composer to Astronomer) and building new remote execution/deployment workflows with robust CI/CD pipelines.
- Delivered foundational data infrastructure for advanced AI/ML use cases, including architecting BigQuery + Vertex AI batch prediction flows and managing technical integration with a 3rd party AI provider.

Amazon, Seattle, WA

May 2023 – August 2023

Software Development Engineer Intern

- Developed a high-availability backend feature for Amazon Photos using C#/.NET, integrating REST endpoints with existing services and data logic.
- Defined and authored the technical design document for the feature under ambiguous requirements, aligning cross-functional teams (PM, SDEs) on the final architecture.
- Optimized data retrieval by engineering SQL database schemas and queries to efficiently handle metadata storage and retrieval via pagination.

Stanford NLP Research, Stanford, CA

November 2023 – February 2024

Graduate AI Research Engineer - Sequence-to-Sequence Semantic Parsing over Wikidata

- Developed a few-shot sequence-to-sequence semantic parser with RAG, enabling natural language queries to generate accurate SPARQL queries. Enhanced entity-linking pipeline by 20%, leading to increased performance over the prior state-of-the-art.

Active Robotic Sensing Laboratory, Raleigh, NC

August 2021 – May 2022

Machine Learning & Robotics Research Engineer

- Led development on constructing the imaging system for the BenchBot, a robotics device to estimate biomass.
- Designed and implemented a robust script to interface with DepthAI cameras and servos, capturing high-quality plant images to train machine learning models.

TECHNICAL SKILLS

Programming Languages

Python, Go, C++, C#, TypeScript

Cloud & Infrastructure

GCP, Terraform, Docker, Kubernetes, Dataproc, Datadog, Pub/Sub

ML/AI

PyTorch, TensorFlow, NumPy, AnyScale, Vertex AI

Data Engineering

Apache Airflow, dbt, BigQuery, SQL